

Motivation

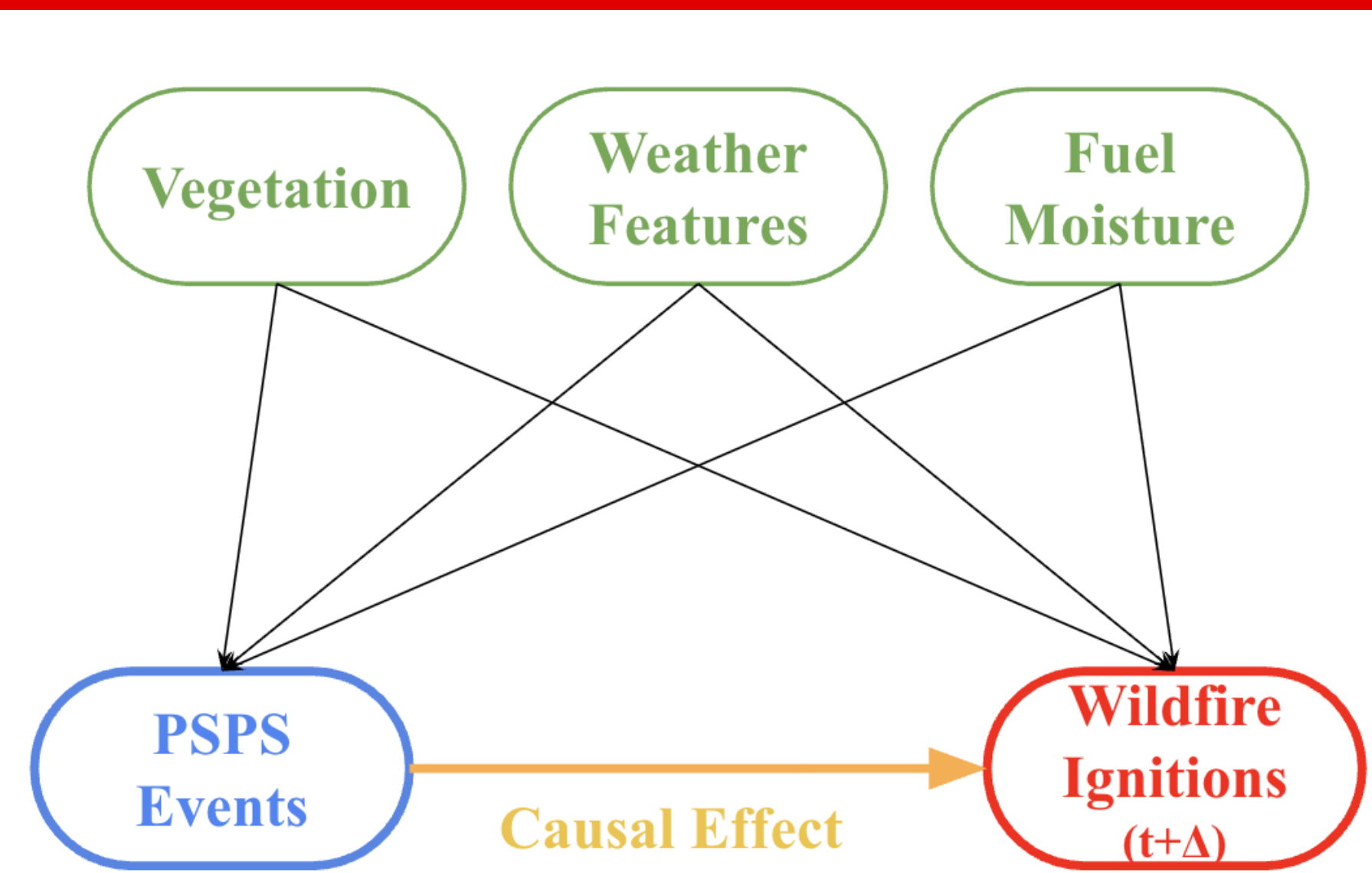
A **Public Safety Power Shutoff** (PSPS) is a proactive and intentional de-energization (shutoff) of electric power lines by utility companies. In regard to this restrictive method, the following high-stakes tradeoff and research gap motivate our research:

- **The High-Stakes Tradeoff:** Utilities use Public Safety Power Shutoffs (PSPS) to prevent wildfires, but this strategy imposes severe societal disruptions and high costs. The core problem: their true effectiveness is unknown.
- **The Critical Gap:** Prior evaluations primarily rely on natural experiments with unrealistic assumptions and limited external validity.
- **Our approach:** We develop a spatio-causal inference framework that integrates operational, weather, and ignition data. This provides the first rigorous, policy-ready evaluation of PSPS impacts to guide resilient mitigation strategies.

The Proposed Framework

1. Model **treatment assignment** mechanism
2. Design **treatment** of interest
3. Estimate the **counterfactual outcomes**

Spatio Causal Estimation



PSPS and Environmental Data in California, 2024

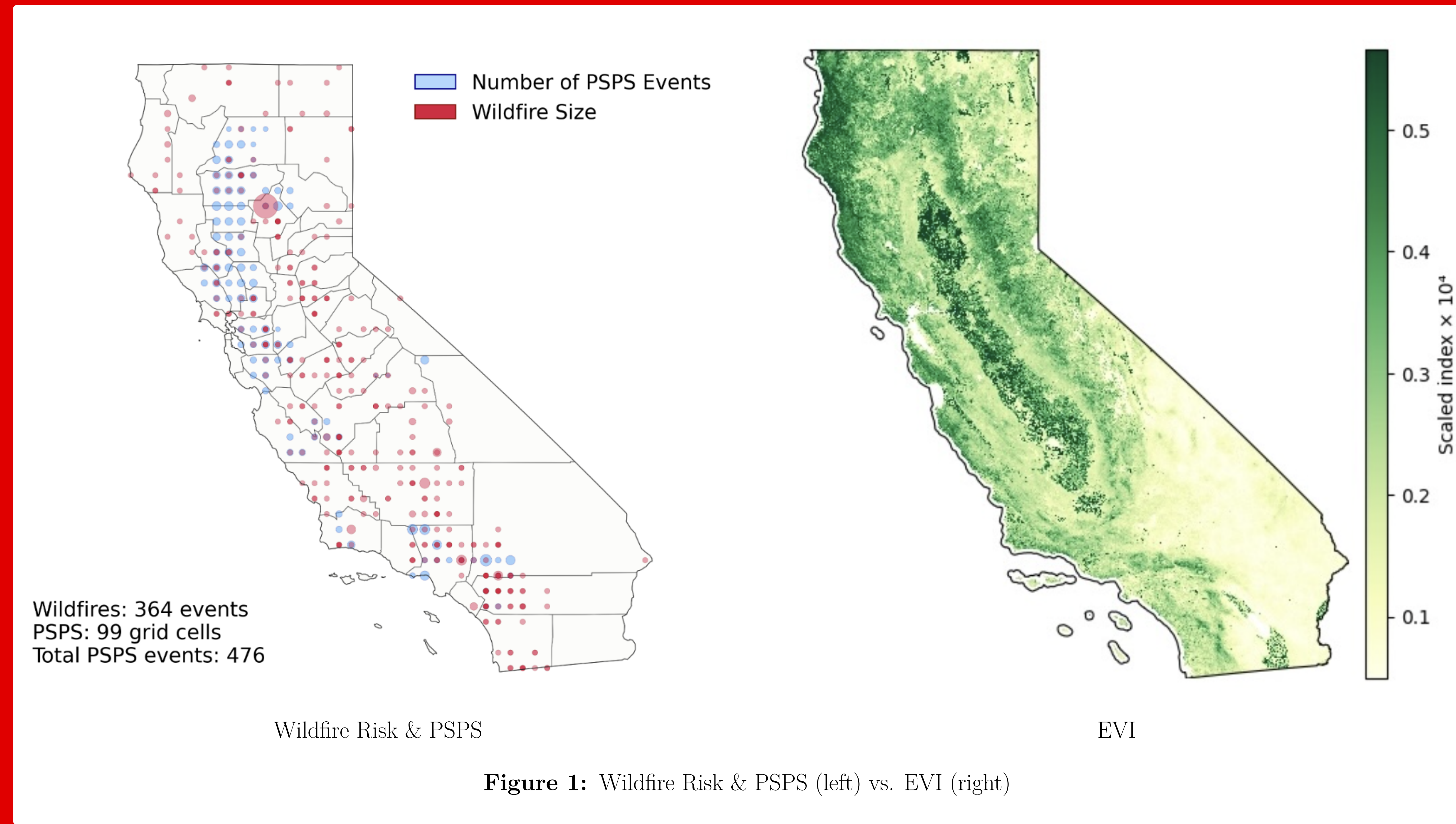


Figure 1: Wildfire Risk & PSPS (left) vs. EVI (right)

Temporal Distribution of PSPS Effects

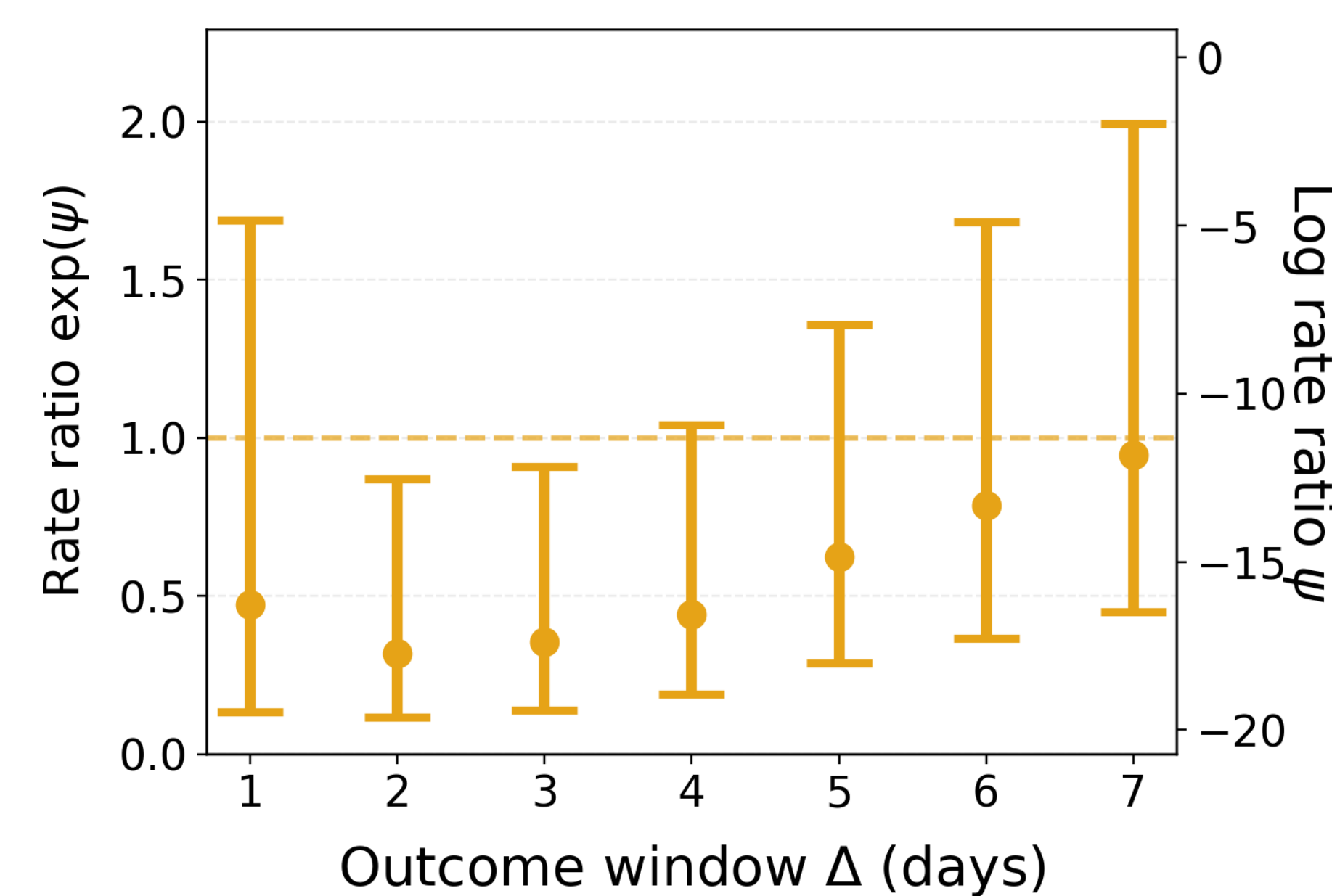


Figure 2: PSPS Effect by Window (With 90% CI)

Takeaway: PSPS reduces ignition risk in the first three days, but estimates are not statistically significant after four days at 10% significance level.

Spatial Distribution of PSPS Effects

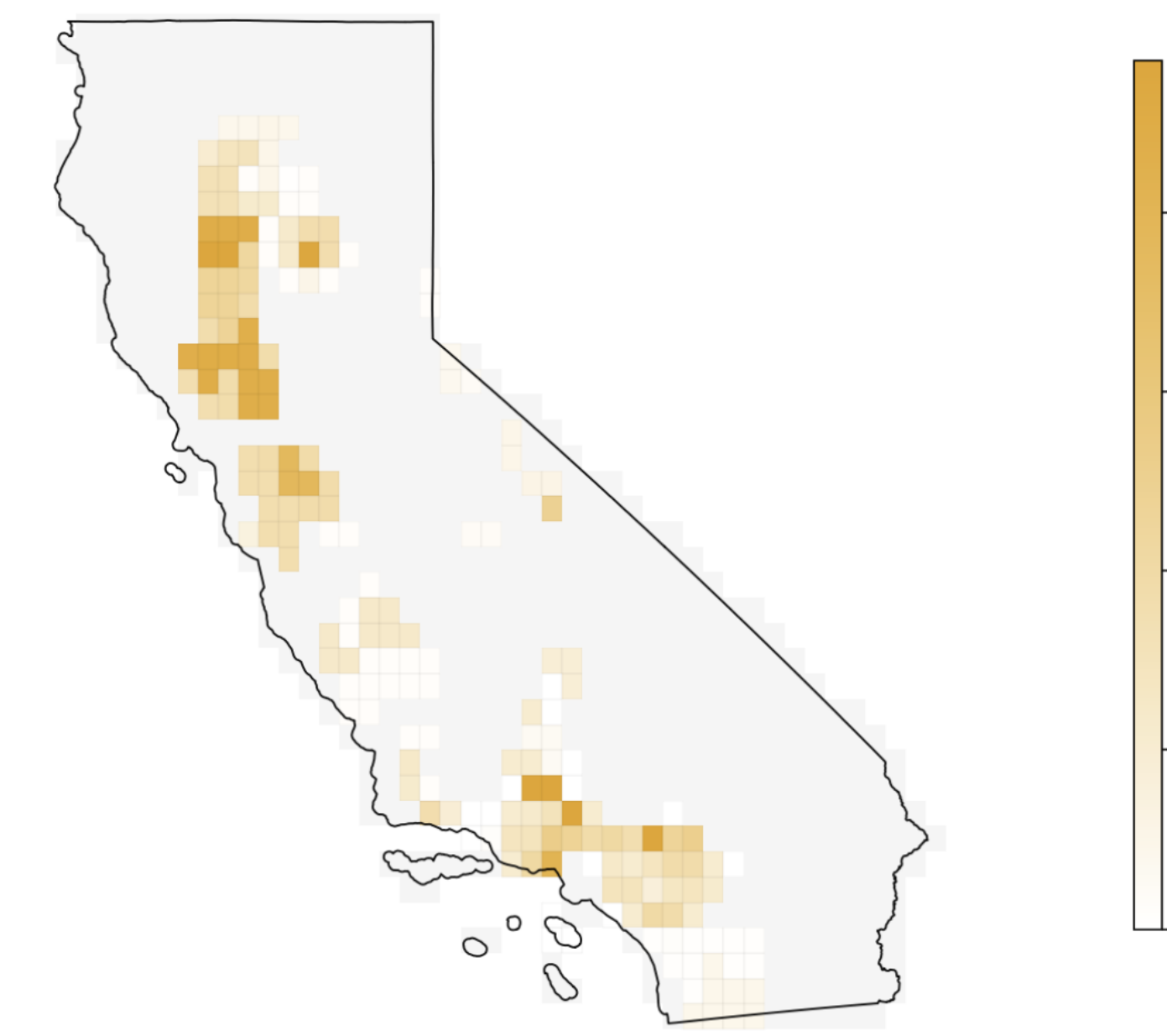


Figure 3: Predicted Ignitions Prevented by PSPS

Takeaway: The map shows percentage of predicted ignitions prevented by PSPS in PSPS-eligible grids, with darker grid cells indicating larger values on the color scale.

Method and Results

Baseline ignition risk model

$$\log \bar{\lambda}_{gt} = s_0(t) + f_{sp}(g) + f_W(W_{gt}) + \sum_{\tau=0}^L \sum_h \alpha_{gh\tau} k(W_{h,t-\tau}, \tau, t). \quad (1)$$

Stabilized inverse-probability weights

$$SW_{gt} = \frac{\tilde{e}_{gt}^{A_{gt}} (1 - \tilde{e}_{gt})^{1-A_{gt}}}{e_{gt}^{A_{gt}} (1 - e_{gt})^{1-A_{gt}}}. \quad (2)$$

Marginal structural outcome model

$$\log \mathbb{E}[N_{g,t+\Delta}] = \alpha_{yw(t)} + \psi A_{gt} + \delta \text{NbrPSPS}_{g,t-1} + \log(B_g \Delta). \quad (3)$$

Notes:

- $A_{gt} \in \{0, 1\}$: PSPS at time t .
- $\text{NbrPSPS}_{g,t-1} = \sum_h G_{gh} A_{h,t-1}$: neighbors' PSPS at $t-1$.
- $\log(B_g \Delta)$: exposure offset (burnable area $\times$ window).
- $\alpha_{yw(t)}$: year-week fixed effects.
- ψ : identified from within-year-week treated vs. untreated contrasts.

Table 1: MSM results across outcome windows Δ (days).

Δ (days)	Effect				Diagnostics
	$\hat{\psi}$ (log RR)	SE($\hat{\psi}$)	RR = exp($\hat{\psi}$)	95% CI	LogLik
1	-0.75	0.77	0.47	[0.104, 2.153]	-3319.20
2	-1.14	0.61	0.32	[0.097, 1.053]	-5922.80
3	-1.03	0.57	0.36	[0.116, 1.089]	-8284.80
4	-0.81	0.52	0.44	[0.160, 1.226]	-10,476.00
5	-0.47	0.47	0.63	[0.249, 1.574]	-12,532.00
6	-0.24	0.46	0.79	[0.318, 1.947]	-14,496.00
7	-0.06	0.45	0.95	[0.390, 2.298]	-16,386.00

Note: PSPS coefficient in MSM Poisson (log link) with offset $\log(B_g \Delta)$.

Takeaway: Across Δ, PSPS shows RR<1 for Δ = 1–3, 0.95 by Δ = 7, suggesting a possible reduction that fades with time. However, all 95% CI crosses 1 and SEs remain large.

Contribution

- **Ignition risk modeling:** Spatio-temporal baseline hazard model combining weather, geography, and lagged neighborhood effects.
- **Causal identification:** Stabilized inverse-probability weighting to address time-varying PSPS assignment.
- **Impact estimation:** Marginal structural outcome model quantifying PSPS effects on ignition risk over time.